

Li-Ion Battery Protection & Voltage Monitoring PCB

Engineering Design Report

Author

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Abstract

This project presents the design and implementation of a compact printed circuit board (PCB) for protecting and monitoring a single-cell 3.7 V Lithium-Ion battery. The circuit incorporates overcurrent protection using a resettable PTC fuse, reverse-current protection using a Schottky diode, battery voltage monitoring through a resistor divider network, and an LED power indicator. The PCB was designed using KiCad and verified through Design Rule Check (DRC) before generating manufacturing-ready Gerber files.

1. Introduction

Lithium-Ion batteries are widely used in embedded systems, IoT devices, robotics, portable electronics, and battery-powered sensors. Improper handling of battery power can lead to reverse current, excessive current flow, or inaccurate battery voltage measurement.

This project demonstrates the implementation of a simple protection and monitoring circuit suitable for educational purposes and embedded system development.

2. Objectives

The primary objectives of this project are:

- Design a manufacturable PCB using KiCad.
 - Protect the battery using a resettable fuse.
 - Prevent reverse current.
 - Provide a clean analog voltage output.
 - Provide easy testing through test points.
 - Generate professional manufacturing files.
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3. System Overview

The PCB contains four major functional blocks:

Battery Input

Receives power from a single-cell Li-Ion battery through an 18650 battery holder.

Protection Circuit

The battery output first passes through

- Resettable 500 mA PTC Fuse
- 1N5819 Schottky Diode

These components provide overcurrent protection and reverse-current protection.

Voltage Monitoring Circuit

The battery voltage is divided by

- $R2 = 100\text{ k}\Omega$
- $R3 = 100\text{ k}\Omega$

The divider output is filtered by

- $C1 = 100\text{ nF}$

The filtered output is available at BAT_ADC.

Output Section

The protected battery voltage is available at connector J1.

A green LED indicates the presence of output voltage.

4. Circuit Operation

When the battery is connected:

1. Current flows through the resettable PTC fuse.
 2. The Schottky diode blocks reverse current.
 3. The protected voltage is available at BAT_OUT.
 4. The LED indicates output power.
 5. The resistor divider scales battery voltage by 2.
 6. The capacitor filters switching noise before the ADC.
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5. Component Selection

Component Purpose

PTC Fuse	Overcurrent Protection
1N5819	Reverse Current Protection
330 Ω	LED Current Limiting
LED	Power Indicator

Component Purpose

100 kΩ	Voltage Divider
100 nF	ADC Noise Filter
Test Points	Debugging

6. PCB Design

The PCB was designed using KiCad.

Key design decisions include

- 90 × 40 mm board size
 - Two-layer PCB
 - Wide power traces
 - Narrow signal traces
 - Large ground plane
 - Compact component placement
 - Easy routing
 - Minimal vias
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7. PCB Layout

(Insert PCB Screenshot)

Describe

- Battery holder placement
 - Power routing
 - Voltage divider placement
 - Test points
 - Ground pour
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8. 3D PCB View

(Insert 3D Image)

Explain that the 3D view verifies component orientation and mechanical fit.

9. Design Rule Check

The PCB passed KiCad Design Rule Check with

Errors: 0

Warnings: 0

This confirms that the PCB is ready for fabrication.

(Insert DRC Screenshot)

10. Gerber Verification

Gerber files were generated and verified using KiCad Gerber Viewer.

Generated layers include

- Top Copper
- Bottom Copper
- Solder Mask
- Silkscreen
- Edge Cuts
- Drill Files

(Insert Gerber Screenshot)

11. Bill of Materials

Insert your BOM table.

12. Manufacturing

The project is compatible with manufacturers including

- JLCPCB
 - PCBWay
 - Seeed Studio
 - OSH Park
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13. Future Improvements

Possible future enhancements include

- Reverse polarity MOSFET protection
- Battery charging circuit
- Over-voltage protection
- USB-C charging

- Fuel gauge IC
 - Battery balancing
 - Current sensing
 - TVS diode protection
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14. Conclusion

A compact, manufacturable battery protection PCB was successfully designed using KiCad. The design satisfies the intended functional requirements, passes DRC validation, and includes complete manufacturing outputs. The project demonstrates practical PCB design skills, component selection, layout optimization, and preparation of fabrication-ready files.

References

- KiCad Documentation
 - ON Semiconductor 1N5819 Datasheet
 - Littelfuse Resettable PTC Datasheet
 - IPC-2221 PCB Design Standard
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Appendix

Include

- Schematic
- PCB Layout
- Gerber Images
- BOM
- DRC Report